

Chapter 7C

Hemostasis and Thrombosis

Platelets, Coagulation Disorders, DIC, TTP/HUS, HIT, VTE, Anticoagulation, and Blood Product Logic

This chapter is designed as the hemostasis/thrombosis backbone for an internal medicine learner and for a future graph-RAG system. It emphasizes mechanism-rich illness scripts, lab pattern recognition, emergency management, and relationships among platelets, coagulation factors, vascular endothelium, anticoagulants, and transfusion products. The uploaded source section covered thrombocytopenia, ITP, TTP, HIT, Bernard-Soulier syndrome, Glanzmann thrombasthenia, von Willebrand disease, hemophilia A/B, DIC, vitamin K deficiency, and coagulopathy of liver disease; this chapter expands those signals into an original integrated framework.

Why this matters clinically

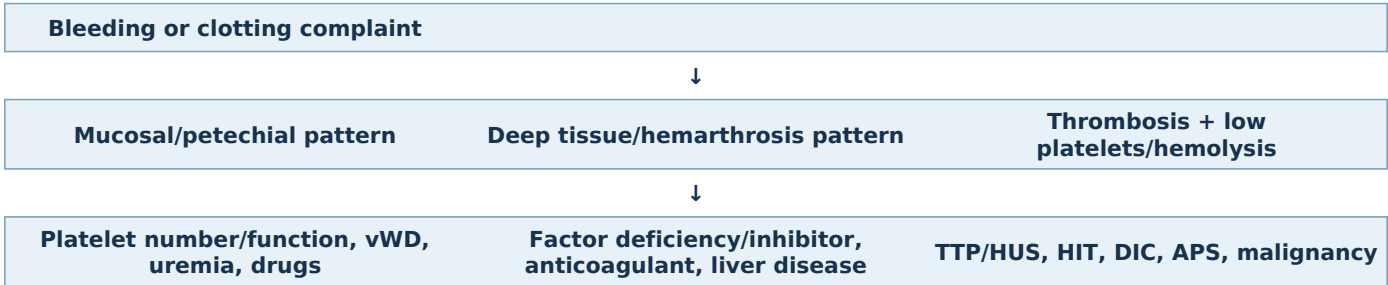
Bleeding and thrombosis are opposite clinical expressions of the same physiologic system. A patient may bleed because platelet number is too low, platelet function is impaired, coagulation factor activity is low, fibrinogen is consumed, fibrinolysis is excessive, blood vessels are abnormal, or anticoagulant medications are too strong. A patient may clot because platelet activation, thrombin generation, venous stasis, endothelial injury, cancer, inflammation, pregnancy, inherited thrombophilia, antiphospholipid antibodies, or heparin-induced platelet activation shifts the system toward thrombosis. The first task is not to memorize every named disorder; it is to ask whether the clinical pattern is mucocutaneous bleeding, deep tissue bleeding, microangiopathic thrombosis, venous thromboembolism, or systemic consumption.

The most dangerous hemostatic diagnoses are time-sensitive. TTP is a plasma-exchange emergency even before ADAMTS13 returns. HIT is a prothrombotic emergency in which giving more heparin worsens the disease. DIC is a syndrome of the underlying illness, so supportive blood products are temporizing while sepsis, malignancy, obstetric catastrophe, trauma, or shock is treated. Massive PE can kill from right ventricular failure before the clot burden is fully characterized. Severe hemophilia bleeding, intracranial bleeding on anticoagulants, and severe thrombocytopenic bleeding all require rapid product- and antidote-based decisions.

Core mental model: primary versus secondary hemostasis

Primary hemostasis begins when vascular injury exposes collagen and von Willebrand factor. Platelets adhere, activate, recruit other platelets, and aggregate into a platelet plug. Secondary hemostasis stabilizes that plug with fibrin through coagulation factor activation and thrombin generation. The clinical distinction is powerful: platelet problems usually cause petechiae, purpura, epistaxis, gingival bleeding, menorrhagia, and immediate oozing after procedures; coagulation factor problems usually cause deep muscle bleeds, hemarthroses, large hematomas, delayed rebleeding after trauma or surgery, and life-threatening internal bleeding.

Original diagnostic orientation map



Question	Primary hemostasis answer	Secondary hemostasis answer	Consumptive/microangiopathic answer
Typical bleeding	Petechiae, purpura, mucosal bleeding, epistaxis, menorrhagia, immediate oozing	Hemarthrosis, muscle hematoma, retroperitoneal bleed, delayed postoperative bleeding	Bleeding plus thrombosis, organ injury, schistocytes, rapidly changing labs
Platelet count	Often low in thrombocytopenia; normal in qualitative platelet defects and vWD	Usually normal unless another process coexists	Often low in DIC, TTP, HUS, HIT
PT/PTT	Usually normal except vWD can mildly prolong PTT if factor VIII falls	One or both prolonged depending on pathway or drug	DIC prolongs PT/PTT and lowers fibrinogen; TTP/HUS often have normal PT/PTT
Immediate danger	Intracranial or GI bleeding when platelets are extremely low; surgical bleeding	Closed-space bleeding, intracranial bleeding, airway/retroperitoneal bleeding	TTP brain/heart/kidney ischemia; HIT thrombosis; DIC shock and bleeding

Initial evaluation of bleeding

The initial bleeding evaluation is a safety screen before it is a diagnostic exercise. Stabilize airway, breathing, circulation, blood pressure, and source control. Ask about anticoagulants, antiplatelet agents, heparin exposure, liver disease, kidney disease, pregnancy, malignancy, infection, autoimmune disease, family history, prior surgical/dental bleeding, mucosal bleeding, thrombosis, and transfusion history. A medication list is diagnostic: aspirin and NSAIDs impair platelet function; clopidogrel, prasugrel, ticagrelor impair P2Y₁₂ signaling; warfarin reduces vitamin K-dependent factors; unfractionated heparin and direct thrombin inhibitors prolong PTT; DOAC effects may not be well captured by routine PT/PTT.

First-line test	What it answers	Important limitations
CBC with differential and smear	Platelet count; anemia; leukemic process; schistocytes; platelet clumping; pancytopenia	EDTA platelet clumping can falsely lower platelet count; smear interpretation matters
PT/INR	Extrinsic/common pathway; factor VII; warfarin effect; liver/vitamin K/DIC clues	Can be normal in mild factor deficiencies and platelet disorders
aPTT	Intrinsic/common pathway; heparin effect; hemophilia A/B; factor inhibitors; lupus anticoagulant	Normal PTT does not rule out mild hemophilia or vWD
Fibrinogen	Consumption or hypofibrinogenemia; DIC severity; cryoprecipitate need	Fibrinogen is an acute-phase reactant and can be normal early in DIC
D-dimer/fibrin split products	Fibrin formation and breakdown; supports DIC/VTE when context fits	Nonspecific: infection, cancer, pregnancy, trauma, inflammation elevate it
Creatinine, LFTs, bilirubin, LDH, haptoglobin	Uremic platelet dysfunction; liver synthetic failure; hemolysis/TMA	Trend and clinical context often more useful than a single value

Interpreting coagulation screening tests

PT and PTT should be treated as pathway clues, not as diagnoses. An isolated prolonged PT suggests warfarin, vitamin K deficiency, early liver dysfunction, or factor VII deficiency. An isolated prolonged PTT suggests heparin, hemophilia A/B, factor XI deficiency, lupus anticoagulant, or an acquired factor VIII inhibitor. Prolongation of both PT and PTT suggests severe liver disease, DIC, massive transfusion/dilution, severe vitamin K deficiency, common pathway factor deficiency, or combined anticoagulant exposure. Normal PT/PTT in a bleeding patient pushes the workup toward platelets, vWD, factor XIII deficiency, vascular fragility, or local anatomic bleeding.

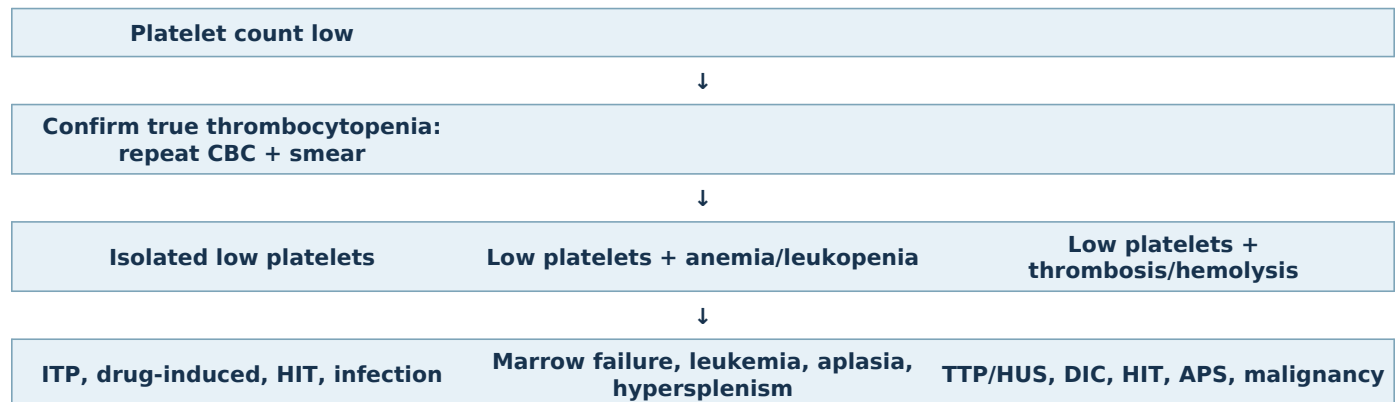
Pattern	Likely diagnoses	Next steps
Normal platelets, normal PT/PTT, mucosal bleeding	vWD, aspirin/NSAIDs/P2Y ₁₂ drugs, uremic platelet dysfunction, inherited platelet function disorder	vWF antigen/activity, factor VIII, platelet function testing if needed, medication review

Pattern	Likely diagnoses	Next steps
Low platelets, normal PT/PTT	ITP, drug-induced thrombocytopenia, HIT, TTP/HUS, hypersplenism, marrow disease	Smear, hemolysis labs, heparin timing, infection/HIV/HCV, bone marrow if production concern
Prolonged PT only	Warfarin, vitamin K deficiency, factor VII deficiency, early liver disease	Medication/diet/antibiotics, liver tests, vitamin K trial when appropriate
Prolonged PTT only	Heparin, hemophilia A/B, vWD with low FVIII, lupus anticoagulant, factor inhibitor	Mixing study, factor assays, lupus anticoagulant testing, anti-Xa if heparin/LMWH suspected
Prolonged PT and PTT + low fibrinogen + low platelets	DIC, severe liver disease with consumption, massive bleeding/transfusion	Treat cause, fibrinogen replacement, platelets/FFP if bleeding/procedure
Schistocytes + thrombocytopenia + hemolysis	TTP/HUS/DIC/malignant HTN/valve hemolysis	Assess PT/PTT/fibrinogen, creatinine, neurologic status, ADAMTS13 if TTP possible

Thrombocytopenia: count, mechanism, and bleeding risk

Thrombocytopenia is a platelet count below the reference range, commonly below 150,000/microliter. The first diagnostic fork is production, destruction/consumption, sequestration, dilution, or pseudo-thrombocytopenia. The second fork is whether the patient is bleeding, clotting, infected, pregnant, exposed to heparin, taking new medications, or has splenomegaly or pancytopenia. Platelet count alone does not determine risk; platelet function, coagulopathy, uremia, sepsis, procedure type, and rapidity of fall all matter.

Original thrombocytopenia algorithm



Platelet range	Common clinical meaning	Usual bleeding risk if platelet function and coagulation are otherwise normal
>100,000/microliter	Often incidental or mild disease; procedures often safe depending on type	Spontaneous bleeding uncommon
50,000-100,000/microliter	Mild to moderate thrombocytopenia; consider trend and procedure risk	Major spontaneous bleeding uncommon; surgical bleeding risk varies
20,000-50,000/microliter	Clinically significant; bruising, petechiae, mucosal bleeding may occur	Bleeding risk increases, especially with trauma/procedures
10,000-20,000/microliter	High-risk zone; evaluate urgently if new or symptomatic	Spontaneous mucosal bleeding may occur
<10,000/microliter	Very severe thrombocytopenia	Highest spontaneous bleeding risk, including intracranial/GI bleeding

Clinical trap: platelet transfusion is not a universal answer to thrombocytopenia. It is lifesaving in hypoproliferative thrombocytopenia with severe bleeding or very low counts, but can worsen thrombosis or fail rapidly in immune destruction. Avoid routine platelet transfusion in suspected TTP or HIT unless there is life-threatening bleeding or an urgent invasive procedure where benefits outweigh risks.

Decreased platelet production

Production failure is suggested by pancytopenia, low reticulocyte count, abnormal white cell differential, macrocytosis, dysplastic smear features, chemotherapy exposure, radiation, alcohol toxicity, viral infections, aplastic anemia, marrow infiltration, leukemia, myelodysplastic syndrome, or nutritional deficiency. A big spleen suggests sequestration, portal hypertension, or hematologic malignancy. Bone marrow biopsy is most useful when thrombocytopenia is unexplained, persistent, associated with other cytopenias, or accompanied by abnormal cells.

Immune thrombocytopenia (ITP)

ITP is an acquired immune thrombocytopenia caused by antibody-mediated platelet destruction and impaired platelet production. The classic adult illness script is an otherwise well patient, often female, with isolated thrombocytopenia, petechiae, epistaxis, or menorrhagia, a normal hemoglobin and white count unless bleeding has occurred, and no splenomegaly. Children often develop acute ITP after viral infection and may recover spontaneously. ITP is a diagnosis of exclusion; do not miss leukemia, marrow failure, DIC, TTP, HIT, drug-induced thrombocytopenia, HIV, HCV, pregnancy-associated thrombocytopenia, or lupus.

ITP domain	High-yield details
Mechanism	IgG-coated platelets are cleared by splenic macrophages; platelet production may also be impaired.
Smear	Reduced platelet number; often large young platelets; no schistocyte-dominant microangiopathy.
Labs	Isolated thrombocytopenia; PT/PTT usually normal. Screen for secondary causes when clinically appropriate.
Treatment logic	Observation if mild and nonbleeding; corticosteroids or IVIG for clinically significant bleeding risk; rituximab, thrombopoietin receptor agonists, or splenectomy for selected persistent/refractory disease.
Product logic	Platelets are reserved for serious bleeding or urgent invasive needs because transfused platelets are also immune-cleared.

Exam clue: severe thrombocytopenia with petechiae but no renal failure, no neurologic changes, no schistocytes, and normal PT/PTT is ITP until proven otherwise. Severe thrombocytopenia plus MAHA is not simple ITP.

Thrombotic microangiopathy: TTP, HUS, and mimics

Thrombotic microangiopathy (TMA) means small-vessel platelet-rich thrombi injure red cells and organs. The shared lab signature is thrombocytopenia plus microangiopathic hemolytic anemia: schistocytes, elevated LDH, indirect hyperbilirubinemia, low haptoglobin, and reticulocytosis. The clinical question is whether the syndrome is immune TTP, Shiga-toxin HUS, complement-mediated atypical HUS, DIC, malignant hypertension, pregnancy-related TMA, drug-related TMA, cancer-associated TMA, or mechanical shear. The presence of thrombocytopenia and schistocytes should trigger urgent action, not a passive wait for every send-out test.

TTP

TTP is caused by severe ADAMTS13 deficiency, most often due to an acquired inhibitor. Without ADAMTS13, ultra-large von Willebrand factor multimers persist, recruit platelets, and produce platelet-rich microthrombi. The classic pentad - thrombocytopenia, MAHA, neurologic findings, renal dysfunction, and fever - is memorable but insensitive. Most patients do not present with all five. Suspect TTP when thrombocytopenia and MAHA occur without another obvious cause, especially when PT/PTT are not markedly prolonged and neurologic symptoms, abdominal pain, cardiac ischemia, or kidney injury are present.

TTP management principle: if clinical suspicion is significant, start therapeutic plasma exchange promptly while sending ADAMTS13 testing. Corticosteroids are typically given for immune TTP; rituximab and caplacizumab may be used depending on severity, relapse risk, and institutional protocols. Do not wait for confirmatory ADAMTS13 when the pretest probability is high.

Feature	TTP	HUS / atypical HUS	DIC
Dominant mechanism	ADAMTS13 deficiency -> ultra-large vWF multimers -> platelet microthrombi	Endothelial injury/complement dysregulation or Shiga toxin -> renal-predominant TMA	Systemic thrombin generation with consumption of platelets/factors/fibrinogen
Platelets	Low, often markedly	Low	Low
Hemolysis	MAHA with schistocytes	MAHA with schistocytes	May have schistocytes from fibrin strands
PT/PTT	Often normal or near normal	Often normal or near normal	Often prolonged
Fibrinogen	Usually not profoundly low	Usually not profoundly low	Often low or falling
Organ clues	Neuro, cardiac, abdominal, renal	Renal prominent; diarrhea prodrome in Shiga-toxin HUS	Shock, sepsis, trauma, obstetric catastrophe, malignancy
Immediate treatment	Plasma exchange + immunosuppression; consider caplacizumab	Supportive; complement blockade for complement-mediated disease	Treat underlying cause + blood products if bleeding/procedure

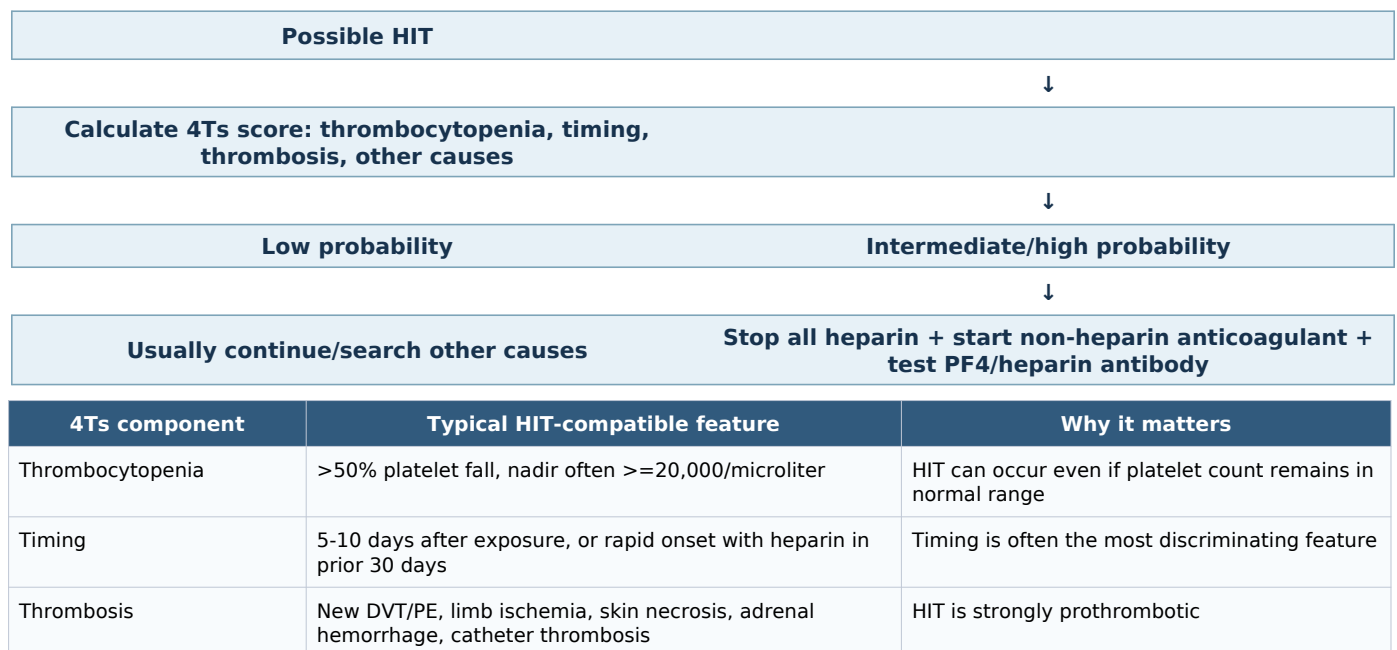
HUS and atypical HUS

Classic HUS often follows Shiga toxin-producing bacterial gastroenteritis and is kidney-predominant. Atypical HUS is commonly complement-mediated and can occur without diarrhea. In adults, TTP and HUS overlap clinically, so immediate management should prioritize excluding or treating TTP when suspicion exists, while also considering complement-mediated disease, malignant hypertension, pregnancy-related TMA, drugs, transplant-associated TMA, and cancer-associated TMA. Severe kidney injury favors HUS but does not exclude TTP.

Heparin-induced thrombocytopenia (HIT)

HIT is an immune, platelet-activating reaction to heparin-platelet factor 4 complexes. It is not primarily a bleeding disorder. It is a thrombosis disorder with falling platelets. The classic timing is a platelet fall 5 to 10 days after heparin exposure, or within 24 hours if there was recent heparin exposure and preformed antibodies. The platelet count typically falls by more than 50% from baseline and may remain above 150,000/microliter, so relying only on an absolute low platelet threshold can miss HIT.

Original HIT diagnostic workflow



4Ts component	Typical HIT-compatible feature	Why it matters
Other causes	Sepsis, DIC, surgery, medications, TMA, splenomegaly, dilution	A plausible alternate cause lowers probability
Treatment: stop unfractionated heparin, low-molecular-weight heparin, heparin flushes, heparin-coated catheters when feasible, and heparin-containing dialysis circuits. Start a non-heparin anticoagulant when clinical probability is not low and bleeding risk permits. Options include argatroban, bivalirudin, fondaparinux, or selected direct oral anticoagulants depending on acuity, renal/hepatic function, procedures, and institutional practice. Warfarin should not be started during acute HIT until platelet recovery because early protein C depletion can worsen thrombosis and skin necrosis.		

Qualitative platelet function disorders

Qualitative platelet disorders present as platelet-type bleeding with normal or mildly decreased platelet counts. The most common acquired causes are aspirin, NSAIDs, P2Y₁₂ inhibitors, uremia, liver disease, cardiopulmonary bypass, myeloproliferative neoplasms, and paraproteinemias. Inherited disorders are rare but high yield: Bernard-Soulier syndrome is defective platelet adhesion due to GPIb-IX-V problems; Glanzmann thrombasthenia is defective platelet aggregation due to GPIIb/IIIa problems.

Disorder	Defect	Platelet count/smea r	Bleeding pattern	Testing clue
Aspirin/NSAIDs	COX inhibition -> reduced thromboxane A ₂ signaling	Normal count	Mucocutaneous bleeding, surgical oozing	Medication history; platelet function abnormal
Uremia	Platelet adhesion/aggregation dysfunction due to uremic toxins	Usually normal count unless other disease	Epistaxis, GI bleeding, oozing from lines	High BUN/ESRD; improves with dialysis, DDAVP, anemia correction
Bernard-Soulier	GPIb-IX-V adhesion defect	Large platelets; mild thrombocytopenia	Mucocutaneous bleeding	Abnormal ristocetin response that does not correct with normal plasma
Glanzmann	GPIIb/IIIa aggregation defect	Normal count and morphology	Mucocutaneous bleeding	Impaired aggregation with multiple agonists; ristocetin response preserved

von Willebrand disease

von Willebrand disease is the most common inherited bleeding disorder. von Willebrand factor has two core jobs: it bridges platelets to injured endothelium through platelet GPIb, and it carries/stabilizes factor VIII. Therefore vWD behaves like a platelet-type mucosal bleeding disorder, but it can also prolong PTT if factor VIII activity is sufficiently reduced. Menorrhagia, epistaxis, easy bruising, gum bleeding, and prolonged bleeding after dental work are common clues; spontaneous hemarthrosis is much more typical of severe hemophilia than routine vWD.

vWD type	Mechanism	Clinical severity	Treatment logic
Type 1	Partial quantitative deficiency of vWF	Most common; usually mild to moderate mucosal bleeding	DDAVP often effective; antifibrinolytics for mucosal procedures; vWF concentrate if severe/procedure or DDAVP nonresponse
Type 2	Qualitative vWF dysfunction	Variable; subtype matters	DDAVP may or may not work; vWF concentrate often used for higher-risk bleeding/procedures

vWD type	Mechanism	Clinical severity	Treatment logic
Type 3	Near-absence of vWF	Severe; can resemble factor VIII deficiency with deep bleeding	vWF-containing factor concentrate; DDAVP ineffective

Testing usually includes vWF antigen, vWF activity, factor VIII activity, and sometimes multimer analysis or platelet-dependent vWF activity assays. Stress, pregnancy, inflammation, estrogen, blood type, age, and thyroid disease can alter vWF levels, so borderline results may require repeat testing in the correct clinical context.

Hemophilia A, hemophilia B, and acquired factor inhibitors

Hemophilia A is factor VIII deficiency and hemophilia B is factor IX deficiency. Both are X-linked and classically affect males, although symptomatic female carriers can occur. The hallmark is deep tissue bleeding: hemarthroses, muscle hematomas, retroperitoneal bleeding, intracranial hemorrhage, and delayed bleeding after trauma, surgery, or dental work. Severe disease presents early in life with spontaneous bleeding; mild disease may first appear after procedures. PTT is prolonged when factor levels are low enough, while PT and platelet count are usually normal.

Severity category	Factor activity	Typical clinical picture
Severe	<1%	Spontaneous hemarthroses and muscle bleeds; early childhood diagnosis; high morbidity without prophylaxis
Moderate	1-5%	Bleeding after minor trauma; occasional spontaneous bleeding
Mild	>5% to <40%	Bleeding mainly after surgery, dental work, major trauma; may be diagnosed late

Hemophilia A treatment includes factor VIII replacement, DDAVP for some mild cases, antifibrinolytics for mucosal bleeding, and increasingly nonfactor prophylaxis such as emicizumab in selected patients. Hemophilia B treatment uses factor IX replacement; DDAVP is not useful. Suspected intracranial bleeding, airway-threatening bleeding, compartment syndrome, iliopsoas bleeding, or major trauma should be treated immediately with factor replacement before imaging when suspicion is high.

Mixing study concept: if prolonged PTT corrects after mixing patient plasma with normal plasma, a factor deficiency is more likely. If it does not correct or correction is time-dependent, an inhibitor is more likely. Acquired factor VIII inhibitor can cause severe bleeding in older adults, postpartum patients, autoimmune disease, malignancy, or idiopathic settings and requires factor-bypassing or porcine factor strategies plus immunosuppression.

Disseminated intravascular coagulation (DIC)

DIC is systemic activation of coagulation with consumption of platelets, fibrinogen, and clotting factors plus secondary fibrinolysis. It is not a primary diagnosis; it is a complication of another process. Common triggers include sepsis, trauma, burns, malignancy, acute promyelocytic leukemia, obstetric catastrophes, shock, severe transfusion reactions, snake envenomation, and severe pancreatitis. Acute DIC often bleeds; chronic DIC may thrombose. Laboratory findings evolve, so trending is more important than a single snapshot.

DIC feature	Mechanism	Expected direction
Platelets	Consumption in widespread thrombin generation	Low or falling
PT/PTT	Factor consumption and impaired coagulation reserve	Prolonged, often PT first
Fibrinogen	Consumed into fibrin clots; may be masked by acute-phase rise	Low or falling; normal does not exclude early DIC
D-dimer/FDP	Fibrin formation and breakdown	Elevated
Smear	Mechanical RBC shearing through microthrombi	Schistocytes may be present

DIC feature	Mechanism	Expected direction
Clinical pattern	Bleeding, thrombosis, organ dysfunction	Line oozing, purpura, ischemic digits, kidney/CNS injury

Treat the underlying cause first. Blood products are supportive and generally directed by bleeding, procedure needs, or severe laboratory derangements. Platelets are used for significant bleeding or high-risk procedures with thrombocytopenia. Cryoprecipitate or fibrinogen concentrate addresses low fibrinogen. FFP replaces multiple clotting factors when bleeding or urgent procedure exists. Anticoagulation may be considered in selected chronic or thrombosis-dominant DIC, but it is not the default response to an actively bleeding patient.

Vitamin K deficiency, warfarin effect, and liver disease coagulopathy

Vitamin K is required to gamma-carboxylate factors II, VII, IX, X and proteins C and S. Factor VII has the shortest half-life, so PT/INR prolongs first. Deficiency occurs with poor intake, prolonged antibiotics that reduce gut flora, fat malabsorption, cholestasis, bariatric surgery, total parenteral nutrition without supplementation, critical illness, and warfarin therapy. PTT prolongs later when deficiency is severe. Platelet count and fibrinogen are usually preserved unless another process is present.

Condition	PT/INR	PTT	Platelets	Fibrinogen	Key clue
Vitamin K deficiency	High early	Normal or high if severe	Normal	Normal	Malnutrition, antibiotics, cholestasis; responds to vitamin K
Warfarin effect	High	Normal or high	Normal	Normal	Medication exposure; reversal depends on bleeding urgency
Liver disease	High, often PT dominant	May be high	Often low from portal HTN/spleno megaly	Normal, low, or dysfibrinogenemia	Low albumin, high bilirubin, cirrhosis signs; both bleeding and clotting risk
DIC	High	High	Low	Low/falling	Shock/sepsis/trauma/cancer; high D-dimer, schistocytes

Cirrhosis creates a “rebalanced” but fragile hemostatic state: procoagulant factors fall, anticoagulant proteins also fall, platelets are often low, fibrinolysis may change, and portal hypertension creates anatomic bleeding risk. An elevated INR in cirrhosis does not mean the patient is auto-anticoagulated against DVT/PE. Hospitalized cirrhotic patients can still need pharmacologic VTE prophylaxis when bleeding risk permits.

Anticoagulants and reversal logic

A bleeding patient on anticoagulation requires four simultaneous questions: What drug is present? When was the last dose? How severe is the bleed or procedure urgency? What is the patient’s renal/hepatic function? Routine PT/PTT can be misleading for DOACs; a normal test does not reliably exclude clinically relevant drug levels for every agent. Use drug-specific assays when available and clinically necessary.

Agent/class	Mechanism	Lab clue	Emergency reversal/support
Warfarin	Reduces vitamin K-dependent factor activation	High PT/INR	Vitamin K for sustained reversal; 4-factor PCC for life-threatening bleeding/urgent reversal; FFP if PCC unavailable
Unfractionated heparin	Antithrombin-mediated thrombin/Xa inhibition	High PTT; anti-Xa useful	Stop drug; protamine reverses rapidly
LMWH	Mostly Xa inhibition through antithrombin	Anti-Xa if needed; PTT often not useful	Stop drug; protamine partially reverses
Dabigatran	Direct thrombin inhibitor	Thrombin time very sensitive; aPTT may rise	Idarucizumab for life-threatening bleeding/urgent surgery; dialysis possible due to renal clearance

Agent/class	Mechanism	Lab clue	Emergency reversal/support
Apixaban/rivaroxaban/edoxaban	Direct factor Xa inhibitors	Anti-Xa calibrated assay if available; PT variable	Andexanet alfa in selected severe bleeding; 4-factor PCC often used when andexanet unavailable or not appropriate
Aspirin/P2Y12 inhibitors	Platelet activation inhibition	Platelet count normal; function impaired	Hold drug; local control; platelet transfusion or DDAVP may be considered in selected severe bleeding contexts

Venous thromboembolism and thrombophilia framework

VTE includes deep vein thrombosis and pulmonary embolism. Mechanistically, VTE reflects Virchow triad: stasis, endothelial injury, and hypercoagulability. The clinical goal is to distinguish provoked from unprovoked events, identify life-threatening PE, determine bleeding risk, choose an anticoagulant, and decide duration. Provoked VTE follows major transient risk factors such as surgery, trauma, hospitalization, estrogen therapy, pregnancy/postpartum period, long travel/immobility, central venous catheters, or acute medical illness. Persistent risk factors include active cancer, antiphospholipid syndrome, inflammatory disease, chronic immobility, and recurrent unprovoked VTE.

Clinical situation	Typical anticoagulation approach	Important caveat
Provoked proximal DVT/PE by major transient risk factor	Treatment phase commonly 3 months	Confirm risk factor truly transient and resolved
Unprovoked VTE	At least treatment phase; consider extended therapy if recurrence risk exceeds bleeding risk	Shared decision-making and bleeding risk assessment matter
Cancer-associated thrombosis	DOAC or LMWH depending on cancer type, bleeding risk, interactions, renal function	GI/GU cancers may have higher bleeding with some DOACs
Pregnancy-associated VTE	LMWH preferred	Warfarin is teratogenic; DOACs generally avoided
Antiphospholipid syndrome	Warfarin often favored, especially triple-positive or arterial events	DOACs can be inferior in high-risk APS
HIT-associated thrombosis	Non-heparin anticoagulant; later transition after platelet recovery as appropriate	Avoid heparin and early warfarin

Inherited thrombophilia testing is often overused. Testing rarely changes management in a patient with a clearly provoked VTE, and acute thrombosis or anticoagulation can distort protein C, protein S, and antithrombin levels. Testing may be considered when results would change duration, family counseling, pregnancy/estrogen decisions, or when there is young age, recurrent VTE, unusual sites, strong family history, or suspicion for antiphospholipid syndrome.

Blood products and procedural thresholds

Blood product choice should match the missing hemostatic component. Platelets replace platelet number and function. FFP replaces multiple soluble coagulation factors. Cryoprecipitate replaces fibrinogen, factor VIII, factor XIII, and vWF-rich proteins, but in modern practice it is most often used for fibrinogen replacement. Prothrombin complex concentrate provides vitamin K-dependent factors for rapid warfarin reversal. Packed RBCs restore oxygen-carrying capacity, not hemostasis.

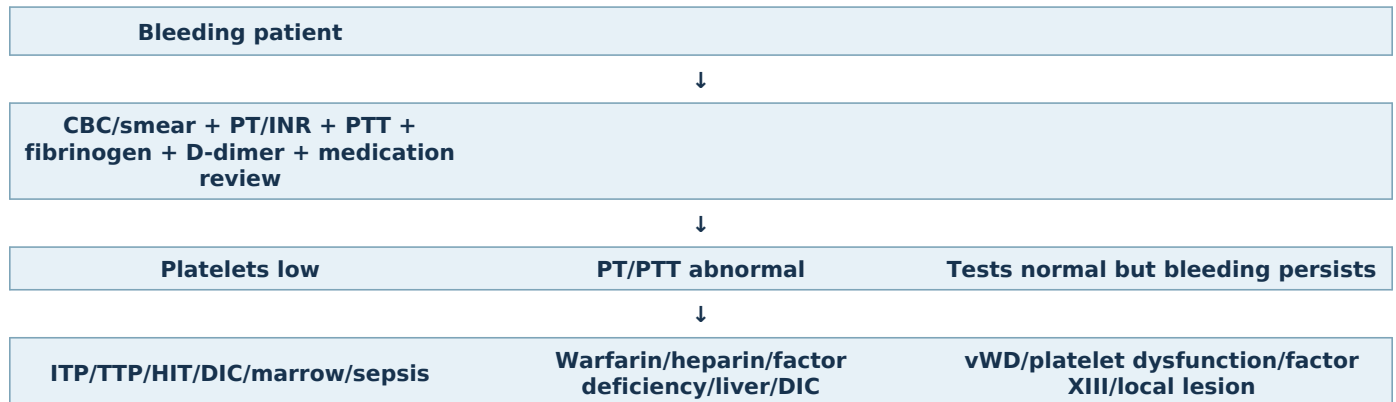
Product	What it contains	Typical hemostatic use	Major cautions
Platelets	Donor platelets	Severe thrombocytopenia with bleeding; prophylaxis in selected very low counts; procedure preparation	Avoid routine use in TTP/HIT; refractoriness, alloimmunization, volume, infection risk
FFP	All soluble clotting factors	Multiple factor deficiency with bleeding/procedure; warfarin reversal if PCC unavailable; plasma exchange replacement in TTP	Large volume; slower than PCC for warfarin reversal

Product	What it contains	Typical hemostatic use	Major cautions
Cryoprecipitate	Fibrinogen-rich concentrate plus factor VIII, vWF, XIII	Hypofibrinogenemia in DIC, massive hemorrhage, obstetric bleeding, trauma	Dose guided by fibrinogen target and bleeding context
4-factor PCC	Factors II, VII, IX, X and proteins C/S	Rapid warfarin reversal with vitamin K; selected DOAC bleeding protocols	Thrombotic risk; not a platelet or fibrinogen product
DDAVP	Releases endothelial vWF and factor VIII; improves some platelet dysfunction	Type 1 vWD, mild hemophilia A, uremic bleeding, selected antiplatelet-associated bleeding	Hyponatremia, tachyphylaxis, limited effect in type 3 vWD/hemophilia B

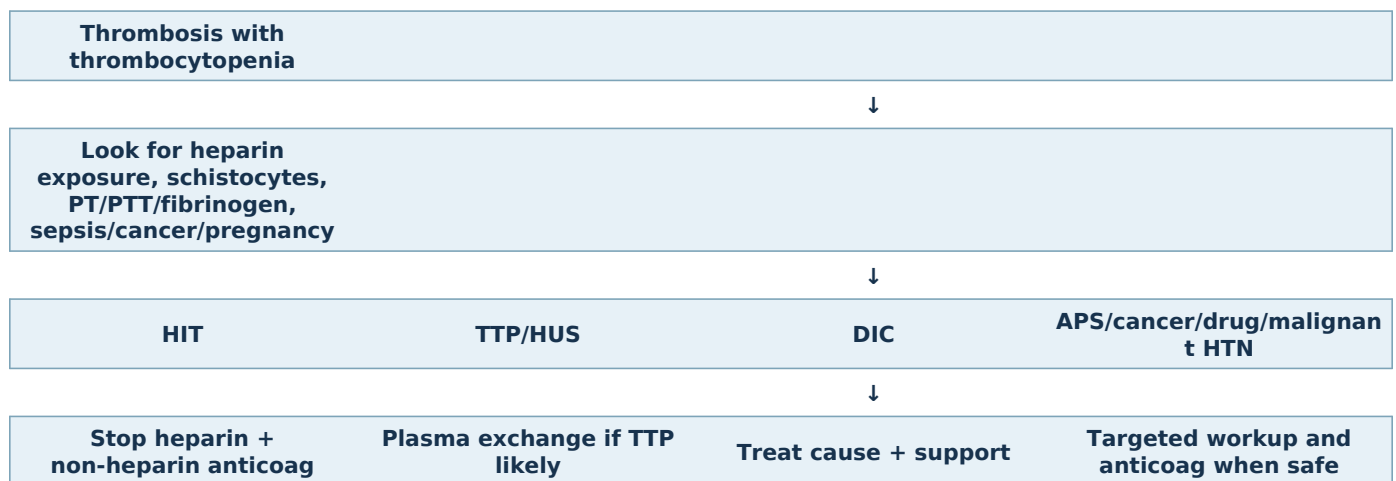
Typical platelet thresholds are context-dependent. A stable nonbleeding patient with therapy-induced hypoproliferative thrombocytopenia is often transfused prophylactically at 10,000/microliter or lower. Higher thresholds are used for active bleeding, major procedures, lumbar puncture, neurosurgery/ocular surgery, or coexisting coagulation defects. Always apply local policy and specialist guidance for procedures because thresholds vary by risk and evidence quality.

Integrated diagnostic algorithms

Bleeding algorithm



Thrombosis + thrombocytopenia algorithm



High-yield disorder comparison table

Disorder	Platelets	PT	PTT	Bleeding/thrombosis signature	Immediate management anchor
ITP	Low	Normal	Normal	Mucosal bleeding, petechiae; isolated thrombocytopenia	Steroids/IVIG if significant; observe mild cases; platelets for severe bleeding
TTP	Low	Normal/near normal	Normal/near normal	MAHA + thrombocytopenia + neuro/cardiac/renal signs	Plasma exchange urgently + steroids; consider caplacizumab/rituximab
HIT	Low or >50% fall	Usually normal	May be normal unless heparin present	Thrombosis after heparin; platelet fall 5-10 days	Stop heparin; non-heparin anticoagulation
DIC	Low	High	High	Bleeding/thrombosis in sepsis, trauma, cancer, obstetric catastrophe	Treat cause; platelets/FFP/cryo if bleeding/procedure
vWD	Normal	Normal	Normal or high	Mucosal bleeding, menorrhagia, dental/surgical bleeding	DDAVP for many type 1; vWF concentrate if severe/nonresponse
Hemophilia A/B	Normal	Normal	High	Hemarthrosis, deep bleeds, delayed bleeding	Factor VIII or IX replacement; DDAVP only some mild A
Vitamin K deficiency	Normal	High first	High if severe	Bleeding with malnutrition, antibiotics, cholestasis	Vitamin K; PCC/FFP if severe bleeding/urgent reversal
Liver disease	Low/normal	High	High/normal	GI/variceal bleeding, thrombocytopenia, fragile hemostasis	Treat source; targeted products; do not assume auto-anticoagulation

Common pitfalls

- Waiting for ADAMTS13 before treating likely TTP. The test is important, but plasma exchange should not be delayed when clinical probability is high.
- Calling every low platelet count ITP. TTP, HIT, DIC, leukemia, aplastic anemia, sepsis, pregnancy-related disorders, and drugs can all present with thrombocytopenia.
- Giving heparin for thrombosis before noticing HIT probability. Thrombosis plus a 5- to 10-day platelet fall after heparin exposure is HIT until proven otherwise.
- Using PT/PTT to rule out DOAC effect. Routine tests may be insensitive or variable for DOACs; timing and renal function are critical.
- Forgetting vWD in menorrhagia or dental bleeding. A normal platelet count and normal PT do not rule out vWD.
- Transfusing platelets reflexively in TTP or HIT. Platelets are generally avoided unless there is severe/life-threatening bleeding or an urgent procedure.
- Assuming cirrhosis protects against thrombosis. Cirrhotic patients can clot and may need VTE prophylaxis despite elevated INR when bleeding risk allows.
- Missing acquired factor VIII inhibitor. New severe bleeding in an older adult or postpartum patient with isolated prolonged PTT should prompt mixing study and factor/inhibitor testing.

Practice questions

Question 1. A 32-year-old woman has petechiae and platelets of 7,000/microliter. Hemoglobin and WBC count are normal, smear has very few platelets and no schistocytes, PT/PTT are normal. What is the most likely diagnosis and initial treatment?

Answer. This is isolated severe thrombocytopenia with platelet-type bleeding and no TMA or coagulation abnormality, most consistent with ITP. If bleeding risk is significant, corticosteroids are first-line and IVIG can be added for rapid platelet rise; platelets are reserved for severe bleeding.

Question 2. A hospitalized patient on unfractionated heparin develops a platelet fall from 280,000 to 110,000/microliter on day 7 and new leg swelling. What should be done before the antibody test returns?

Answer. This is compatible with HIT: timing 5-10 days, >50% fall, and thrombosis. Stop all heparin and start a non-heparin anticoagulant if bleeding risk permits while sending PF4/heparin immunoassay and confirmatory testing as appropriate.

Question 3. A patient has confusion, platelets 18,000/microliter, hemoglobin 7.8 g/dL, schistocytes, high LDH, low haptoglobin, creatinine 1.7 mg/dL, normal PT/PTT. What is the emergency?

Answer. TTP is the emergency. Send ADAMTS13 before treatment if it does not delay care, but start plasma exchange urgently with corticosteroids because untreated TTP can be rapidly fatal.

Question 4. A septic patient has oozing from venipuncture sites, platelets 45,000/microliter, PT/PTT prolonged, fibrinogen low, and D-dimer high. What is the syndrome and main treatment principle?

Answer. This is DIC from sepsis. Treat the underlying infection/shock and support hemostasis with platelets, FFP, and fibrinogen replacement if bleeding or invasive procedures require it.

Question 5. A teenage boy has recurrent knee swelling and prolonged PTT with normal PT and platelets. What diagnosis should be considered?

Answer. Hemophilia A or B should be considered. Order factor VIII and IX activity. Treat serious bleeding with the appropriate factor replacement; DDAVP can help selected mild hemophilia A but not hemophilia B.

Question 6. A woman has lifelong epistaxis and menorrhagia. Platelet count and PT are normal; PTT is mildly prolonged. What diagnosis connects platelet-type bleeding and factor VIII?

Answer. von Willebrand disease. vWF mediates platelet adhesion and stabilizes factor VIII, explaining mucosal bleeding and possible PTT prolongation.

Question 7. A malnourished patient on broad-spectrum antibiotics develops prolonged PT with initially normal PTT, normal platelets, and normal fibrinogen. What is likely?

Answer. Vitamin K deficiency. Factor VII falls first, so PT prolongs early. Treat with vitamin K; use PCC/FFP if urgent bleeding/reversal is needed depending on context.

Question 8. A cirrhotic inpatient with INR 2.0 and platelets 80,000/microliter is immobile after surgery. Is the INR alone proof that pharmacologic VTE prophylaxis is unnecessary?

Answer. No. Cirrhosis is a rebalanced hemostatic state and does not auto-anticoagulate the patient. VTE prophylaxis depends on bleeding risk, platelet count, procedure status, and local guidance.

Graph-RAG extraction map

The following map is designed for concept-node extraction. Each row can become a node, edge, or attribute bundle in a medical graph RAG. Suggested edge types include causes, presents_with, diagnosed_by, treated_by, complicates, mimics, contraindicates, and requires_urgent_action.

Node	Core attributes	High-yield edges
Primary hemostasis	Platelet adhesion/activation/aggregation; vWF; mucocutaneous bleeding	vWD impairs; aspirin impairs; uremia impairs; thrombocytopenia reduces
Secondary hemostasis	Coagulation factors and fibrin stabilization; PT/PTT pathways	Hemophilia impairs; warfarin impairs; liver disease impairs; DIC consumes
Thrombocytopenia	Platelets <150,000/microliter; mechanism production/destruction/sequestration/dilution	ITP causes; HIT causes; DIC causes; TTP causes; marrow failure causes
ITP	Isolated immune platelet destruction; normal PT/PTT; mucosal bleeding	treated_by steroids/IVIG; mimics TTP/HIT; severe_bleeding_requires platelets
TTP	ADAMTS13 deficiency; ultra-large vWF; MAHA; thrombocytopenia; normal PT/PTT	requires_urgent_action plasma exchange; treated_by steroids/caplacizumab/rituximab; mimics HUS/DIC

Node	Core attributes	High-yield edges
HIT	Anti-PF4/heparin platelet activation; thrombosis; platelet fall 5-10 days	caused_by heparin; treated_by non-heparin anticoagulants; contraindicates heparin/early warfarin
DIC	Systemic thrombin generation; consumption; low fibrinogen; high D-dimer	caused_by sepsis/trauma/obstetrics/cancer; treated_by source control + products
vWD	vWF quantity/function defect; mucosal bleeding; possible prolonged PTT	treated_by DDAVP/vWF concentrate; connected_to factor VIII and platelet adhesion
Hemophilia A/B	Factor VIII/IX deficiency; prolonged PTT; hemarthroses	treated_by factor replacement; DDAVP treats mild A; causes deep_bleeding
Vitamin K deficiency	Reduced II,VII,IX,X and proteins C/S; PT first	caused_by antibiotics/malabsorption/warfarin; treated_by vitamin K/PCC/FFP
Liver disease coagulopathy	Reduced factor synthesis, thrombocytopenia, portal HTN; both bleeding and clotting risk	mimics DIC; complicates procedures; does_not_equal auto-anticoagulation
VTE	DVT/PE; Virchow triad; provoked/unprovoked/cancer/pregnancy/APS	treated_by anticoagulation; complicated_by PE/RV failure; risk_increased_by HIT/cancer/APS

References used for clinical cross-checking

- American Society of Hematology. 2018 guidelines for management of venous thromboembolism: heparin-induced thrombocytopenia.
- International Society on Thrombosis and Haemostasis. Guidelines for diagnosis and treatment of thrombotic thrombocytopenic purpura, including updated guidance where available.
- AABB/ICTMG platelet transfusion guideline resources and prior AABB platelet transfusion threshold guidance.
- CHEST antithrombotic therapy for VTE disease guideline updates.
- Standard internal medicine hematology references for vWD, hemophilia, DIC, vitamin K deficiency, and liver disease coagulopathy.